

Niloufar Ebrahimi

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SUMMARY

Data enthusiast with a strong background in statistical analysis and three years of experience in data-driven decision-making within the pharmaceutical industry. Proficient in Python, R, and SQL, with a proven ability to translate complex data into actionable insights and strategic recommendations. Eager to apply my expertise in data analysis, predictive modeling, and cross-functional collaboration to support business initiatives and drive meaningful outcomes.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL/MySQL

Tools: Excel (Pivot Tables, dashboards), Tableau, Metabase

Data Science: Big Data analytics, anomaly detection, predictive modeling, Statistical Inference, A/B testing

Machine Learning: Regression, Classification, Clustering, Dimensionality Reduction (PCA), Neural Networks

Libraries: Pandas, NumPy, SciPy, scikit-learn, caret, mlr

PROFESSIONAL & WORK EXPERIENCE

Statistician | Data Analyst

Medical Department, [Orchid Pharmed Company](#), Tehran, Iran

- Collaborated with cross-functional teams, including executives and clinicians, to define key questions and translate them into data-driven analytical plans that informed strategic decisions.
- Conducted statistical and predictive analyses (R/Python) on large-scale health and customer-related datasets to uncover trends, segment populations, and forecast outcomes.
- Developed predictive models using statistical and machine learning techniques to forecast key performance indicators, resulting in a quantifiable improvement in operational efficiency.
- Developed interactive dashboards and reports to monitor performance, and deliver actionable insights to non-technical stakeholders.
- Optimized relational database queries to enable faster, more reliable reporting and reduce manual reporting time.
- Mentored and supervised a team of three junior analysts, fostering skills in SQL querying, and data-driven reporting.

EDUCATION

M.S. in Mathematical Statistics, Amirkabir University of Technology, Tehran, Iran

2017 - 2021

B.S. in Applied Statistics, University of Tehran, Tehran, Iran

2012 - 2017

RESEARCH EXPERIENCE

Research Associate, lab of [Dr. Adel Mohammadpour](#)

2019 - 2021

School of Mathematics and Computer Science, Amirkabir University of Technology, Iran

- Worked on linear dimensionality reduction methods in high-dimensional settings using R & Python.
- Developed robust regression methods for high-dimensional data using optimization and ML techniques.

Undergraduate Research Assistant, lab of [Dr. Samaneh Eftekhari](#)

2017

School of Mathematics, Statistics and Computer Science, University of Tehran, Iran

- Data acquisition, preprocessing, and statistical modeling for a project on credit risk assessment.
- generate robust predictive insights from real-world datasets.

TEACHING EXPERIENCE

Workshop Instructor in Probability and Statistics for Engineers and Scientists

2018 - 2020

Department of Textile Engineering, Amirkabir University of Technology

- Led three workshops on Statistics for engineering students.

Teaching Assistant, Linear Models course for Graduate students 2019
Department of Mathematics and Computer Science, Amirkabir University of Technology

- Led weekly classes for a graduate-level course, effectively communicating complex quantitative concepts using R to emphasize statistical analysis and problem-solving.

Workshop Instructor in Data Analysis with R Programming Language 2019
Department of Mathematics and Computer Science, Amirkabir University of Technology

- Led workshops on Data Analysis with R, effectively communicating complex programming and statistical concepts to graduate students with varied technical backgrounds.

Undergraduate Teaching Assistant, Sampling Methods course 2016
Department of Mathematics and Computer Science, University of Tehran

PROFESSIONAL DEVELOPMENT & CERTIFICATION

Neural Networks and Deep Learning In Progress
 Coursera Course Certificate

Machine Learning For Healthcare 2024
 MIT OpenCourseWare

Machine Learning internship Summer 2019
School of Computer Science, Institute for Research in Fundamental Sciences (IPM), Iran

Data Science internship Summer 2018
School of Computer Science, Institute for Research in Fundamental Sciences (IPM), Iran

Conference Papers

Ebrahimi, N., & Mohammadpour, A. (2021). Minimum distance Lasso for robust estimation in high-dimensional data. *Journal of Statistical Modeling: Theory and Applications (JSMTA)*, 2(1), 143–147. [\[Link\]](#) 2021

Ebrahimi, N., Vahidi, A., & Salavati, E. (2018). Comparative analysis of machine learning methods for credit risk assessment. *National Conference on Financial and Actuarial Mathematics (FINACT-IRAN)*, 5(1), 32–35. [\[Link\]](#) 2018

Ebrahimi, N., & Eftekhari, S. (2017). Evaluating loan risk determinants with logistic regression. *National Conference of Monetary and Exchange Policies*, 27(1), 181–203. [\[Link\]](#) 2017

Comparative analysis of regression approaches for high-dimensional data. Poster presented at the National Conference on Applied Stochastic Processes, Alzahra University, Iran. 2019

REFERENCES

Dr. Adel Mohammadpour, Associate Professor, Department of Mathematics and Computer Science, Amirkabir University of Technology [Email: adel@aut.ac.ir](mailto:adel@aut.ac.ir) My Master's advisor

Ramin Shahpari, Health Data Science Manager, Department of Medical, Orchid Pharmed [Email: shahpari.r@orchidpharmed.com](mailto:shahpari.r@orchidpharmed.com) My Manager @ Orchid

Mahdiye Bayat, Senior Statistician @ AstraZeneca [Email: Mahdiye.bayat@astrazeneca.com](mailto:Mahdiye.bayat@astrazeneca.com) My Team lead @ Orchid